

Group 8

Real-Time Custom Object Detection and Tracking on
NVIDIA Jetson Orin Nano



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Project Goal



- ▶ Our goal is to build a real-time custom object detection system using NVIDIA Jetson Orin Nano and a webcam.
- ▶ The system should be able to detect and track common objects across frames from live camera input, displaying the object category, location, and movement in real time.

System Approach



- ▶ We collected and organized a custom object dataset for training.
- ▶ For the YOLO dataset, we used six object classes: Soccer Ball, Keyboard, Cellphone, Laptop, Mouse, and Pen.
- ▶ We trained a YOLOv8-based object detection model using this custom dataset.
- ▶ The best-performing model checkpoint was deployed on the NVIDIA Jetson Orin Nano, where live webcam frames are passed through the model to output detected object categories and bounding box locations in real time.

Dataset Overview

- The final YOLO dataset used six object classes with labeled bounding boxes:

Labeled object instances:

Soccer Ball: 875 instances

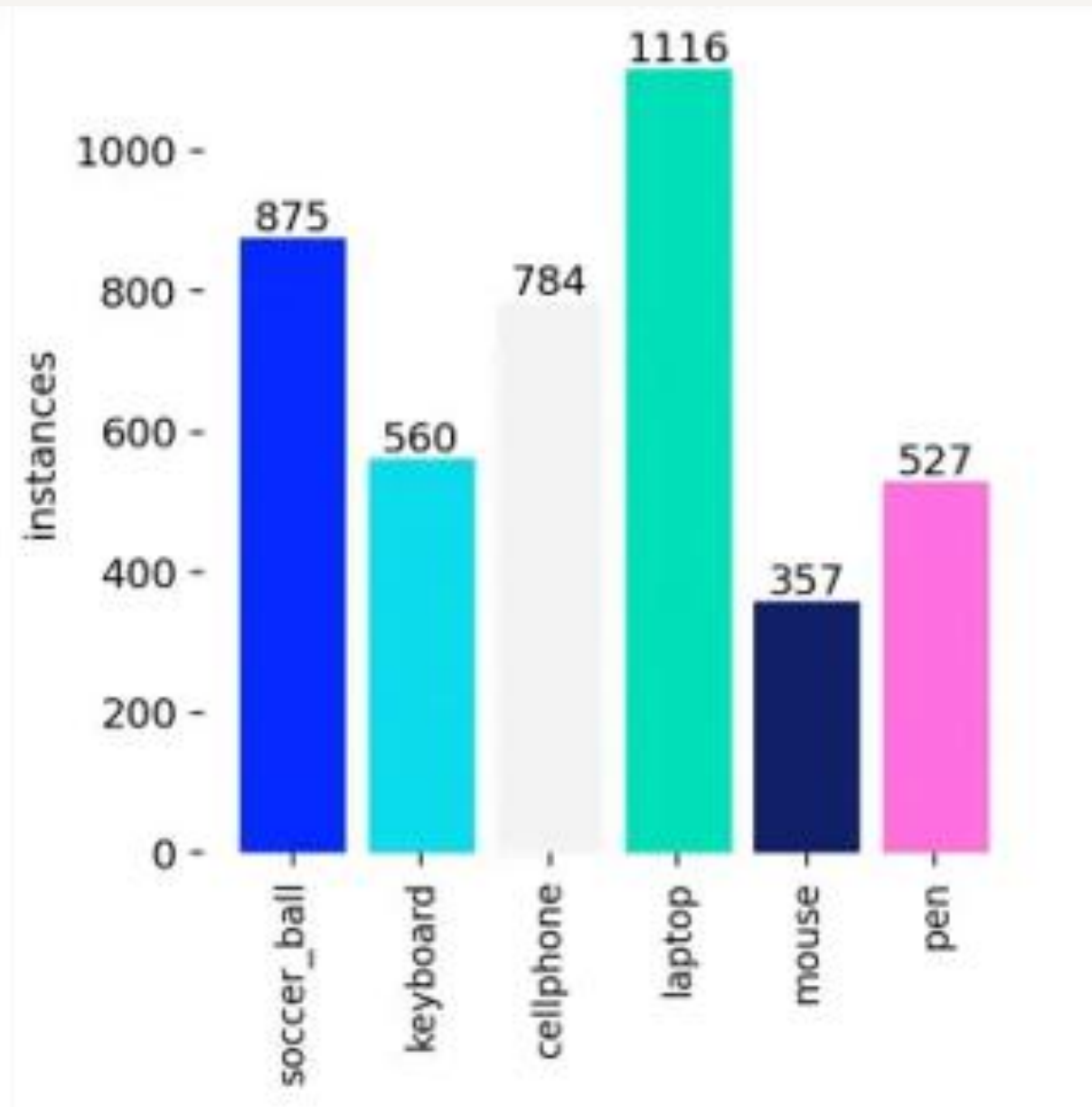
Keyboard: 560 instances

Cellphone: 784 instances

Laptop: 1,116 instances

Mouse: 357 instances

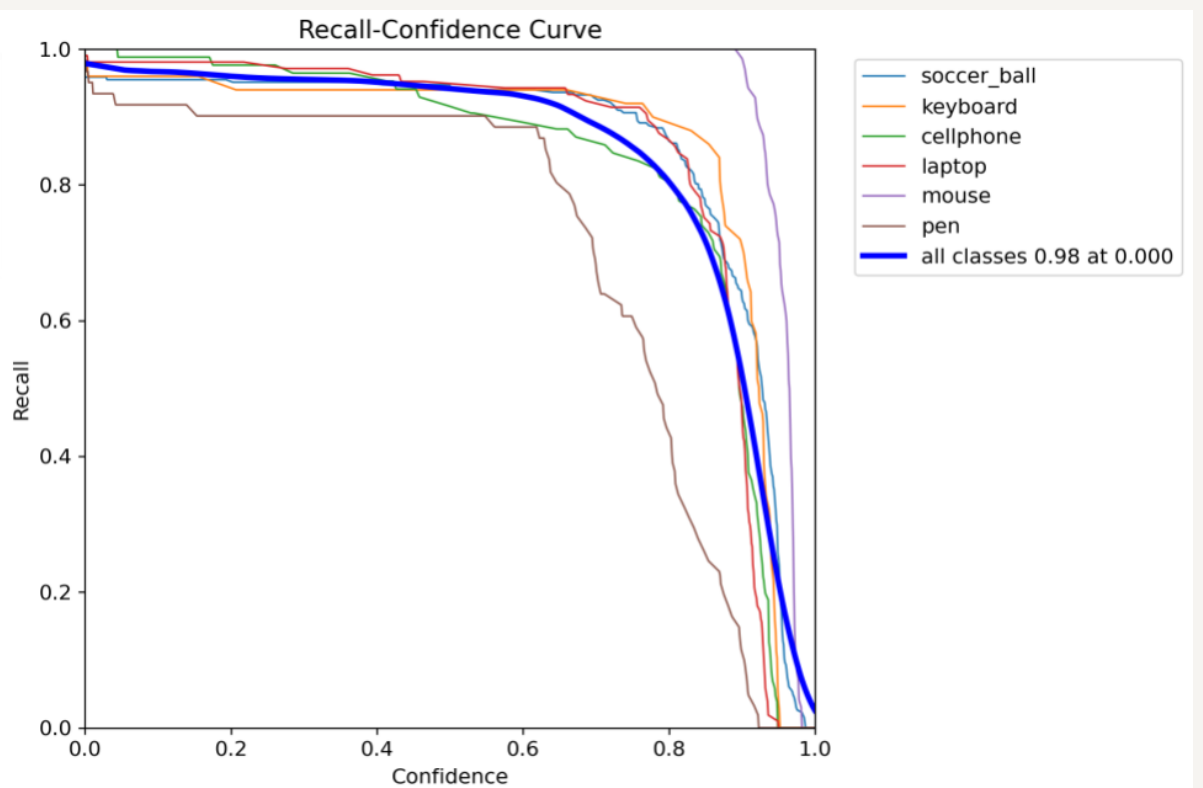
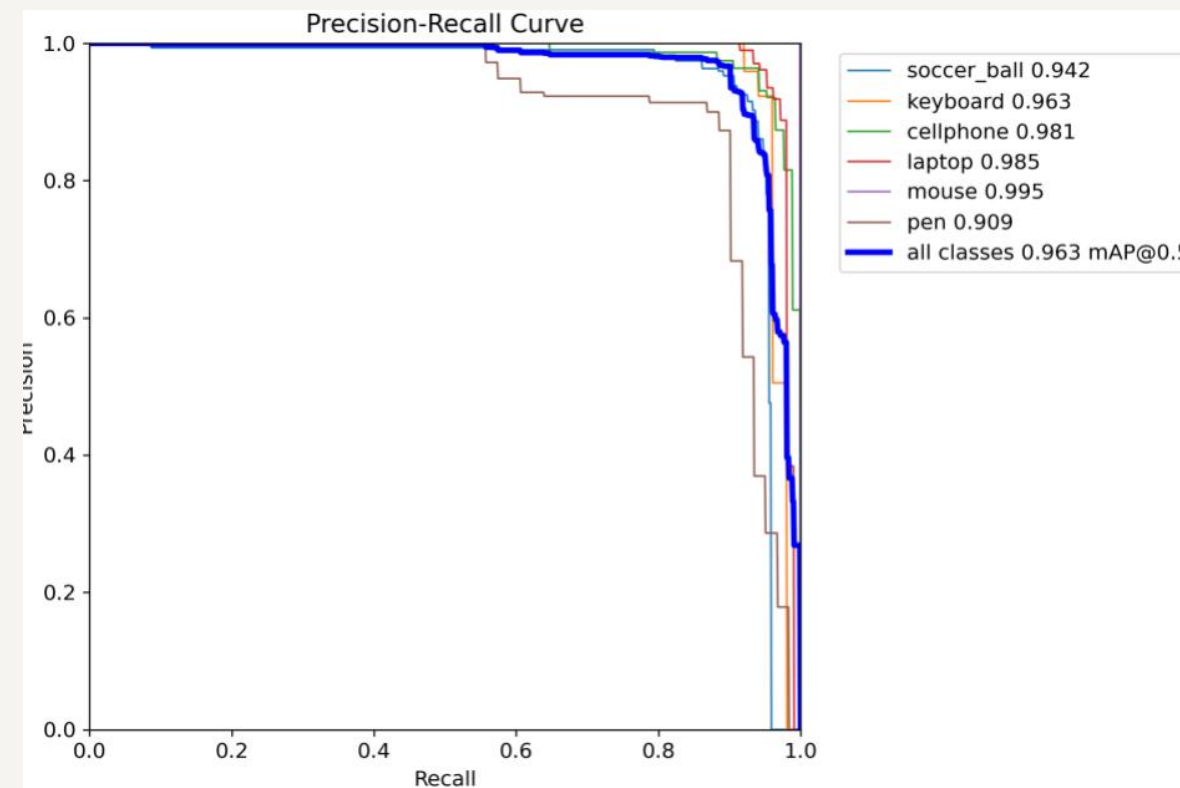
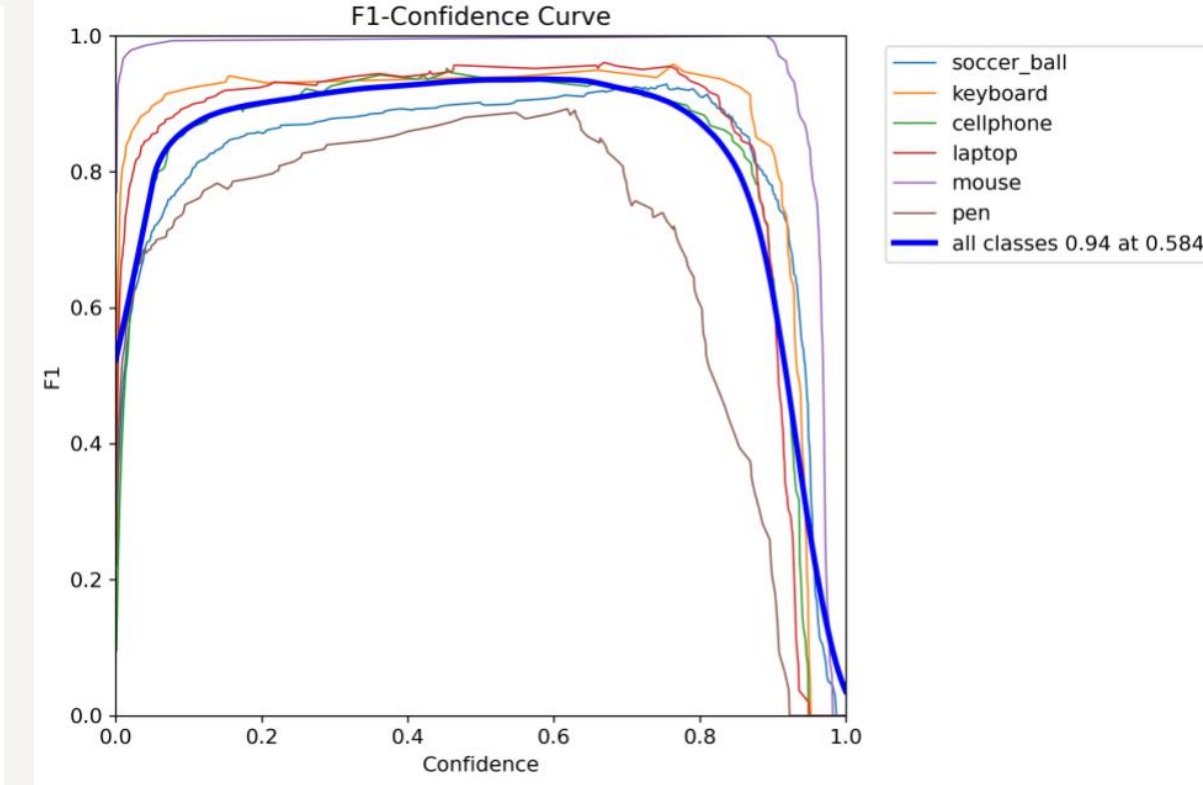
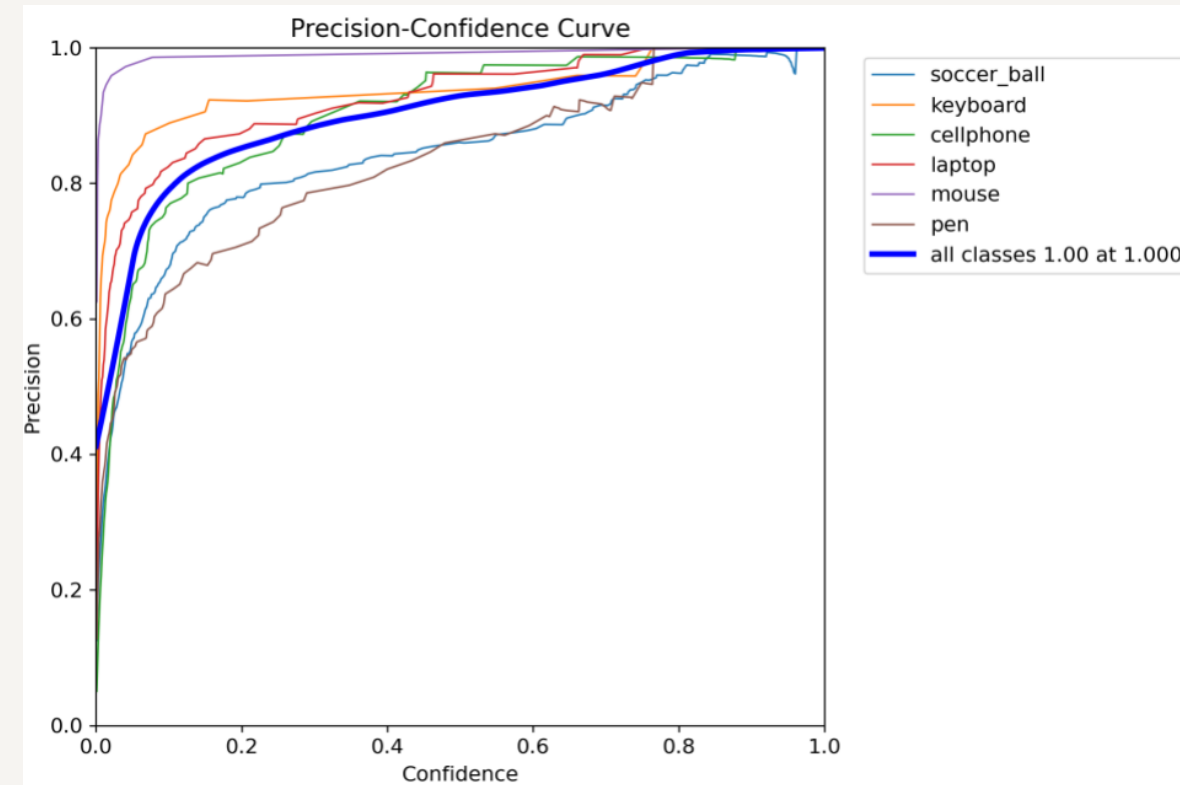
Pen: 527 instances



Model Training Results

Training results showed decreasing losses and improving precision, recall, and mAP scores.

This indicates that the model learned to detect the selected object classes effectively.

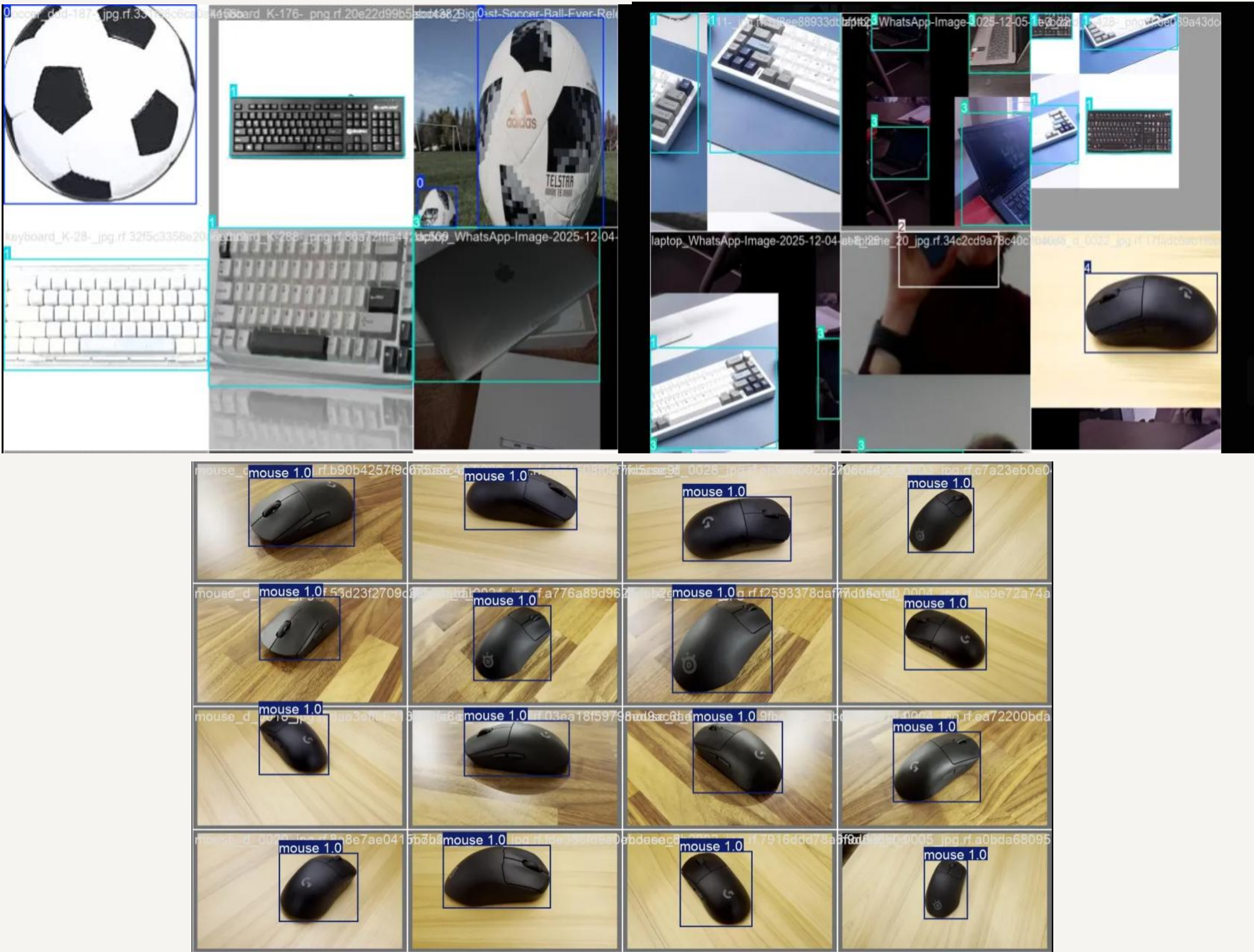


Detection Output Example



The trained YOLO model detects selected objects and displays the predicted class label, confidence score, and bounding box.

The examples show detections for keyboard, soccer ball, laptop, and mouse.



Demo



We will run the trained YOLO model using Jetson Orin Nano and a webcam.

We will place objects in front of the webcam and show how the system detects objects in real time.

The demo will focus on objects from the final custom dataset, such as:
Mouse, Keyboard, Cellphone,
Laptop, Pen, and Soccer Ball

The output screen will display:

- ▶ detected object category
- ▶ model confidence score
- ▶ a box around the detected object

Conclusion



As demonstrated, our YOLOv8-based system can detect selected custom objects in real time using the NVIDIA Jetson Orin Nano and a webcam. The final dataset focuses on six demo-related classes: Soccer Ball, Keyboard, Cellphone, Laptop, Mouse, and Pen. The results show that targeted dataset preparation and YOLO training can support effective real-time custom object detection.